

① Define data mining & list any two basic data mining tasks

A) data mining:- data mining is the process of extracting useful patterns & knowledge from large databases.

Two tasks:-

1. classification, 2. clustering.

② What is data cleaning? mention two common data quality issues.

A) Data cleaning:- It is the process of removing errors & inconsistencies from data.

③ Define data integration and list two challenges.

A) Data integration:- It combines data from multiple sources into a single, consistent database.

Challenges:

1. Data redundancy
2. Data inconsistency.

④ What is covariance analysis? List the two covariance formulas.

A) covariance analysis:- It measures how two variables vary together.

Formulas

1. Population covariance

2. Sample covariance.

⑤ Will the stock prices rise or fall together?
 A) → The correlation coefficient is approximately +0.99, showing a strong positive correlation.
 • The stock prices rise & fall together & are not independent.

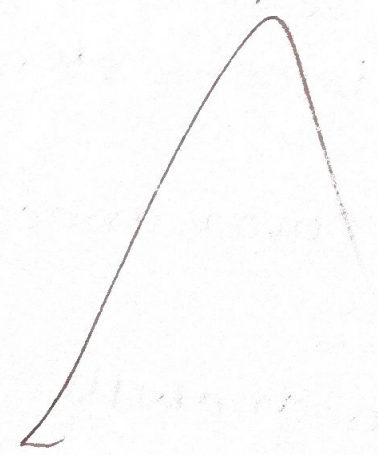
⑥ Explain the architecture of a data mining system.

- A) Components:-
- Data sources
 - Database / Data warehouse
 - Data mining engine
 - Pattern evaluation
 - user interface.

⑦ Find covariance for $x = \{4, 5, 6, 7, 9\}$, $y = \{8, 3, 7, 5, 6\}$.

A) Given:-

- $x = \{4, 5, 6, 7, 9\}$
- $y = \{8, 3, 7, 5, 6\}$
- $\bar{x} = 6.2, \bar{y} = 5.8$
- $\sum (x - \bar{x})(y - \bar{y}) = -1.8$



Sample Covariance:-

$$\text{Cov}(x, y) = \frac{-1.8}{4} = -0.45$$

Qin 3:- 35, 35, 35, 36, 40, 45, 46, 52, 70

mean = 43

Smoothed:- 43, 43, 43, 43, 43, 43, 43, 43.

10. Explain data discretization & its importance.
- A) Data discretization:- It converts continuous data into discrete intervals (bins).
- Importance:- It reduces data complexity & improves the efficiency of data mining algorithms.